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Building an Advanced Interactive Iris Dashboard in Power BI with R Integration

Greetings everyone! Are you interested in finding out how to turn any old dataset (like the Iris flower data classification) into an interactive dashboard? This project illustrates how I built an advanced analytical business report using Power BI and R.

1. Experiment Overview and Purpose

Though the Iris flower data classification was traditionally used for algorithm scripts, in this experiment, we will explore how we can manipulate a scientific data set into building an analytical dashboard. The purpose of this experiment is to:

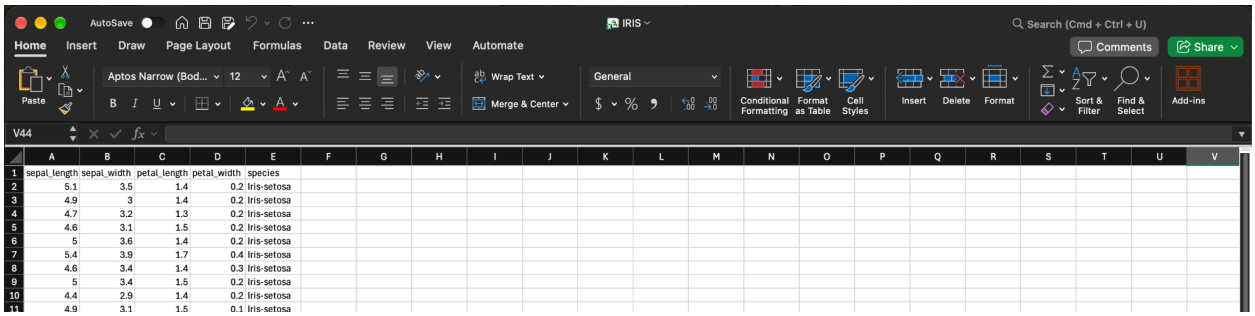
- ➔ Import exploratory data into Power BI
- ➔ Clean and transform the data using query editor and column functions
- ➔ Build DAX expressions
- ➔ Visualize insights in the form of cards, visuals, and R visualizations

Before we start, make sure that you have both Power BI and RStudio installed on your computer.

2. Data Exploration and Preprocessing

The Iris flower data set contains 150 instances of flowers, with 4 physical measurements and 1 target species classification. As a way to prepare our data for analytical processing, we will want to utilize Power Query to both clean and engineer new features from our data. Below are the original and engineered features:

Original Features: sepal_length, sepal_width, petal_length, petal_width, and species

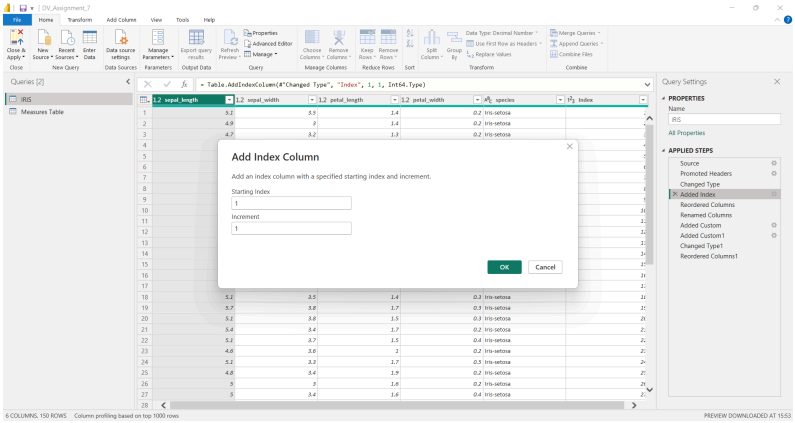


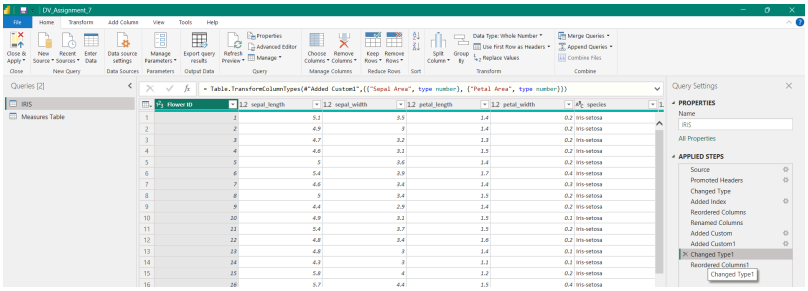
	A	B	C	D	E
1	sepal_length	sepal_width	petal_length	petal_width	species
2	5.1	3.5	1.4	0.2	Iris-setosa
3	4.9	3	1.4	0.2	Iris-setosa
4	4.7	3.2	1.3	0.2	Iris-setosa
5	4.6	3.1	1.5	0.2	Iris-setosa
6	5	3.6	1.4	0.2	Iris-setosa
7	5.4	3.9	1.7	0.4	Iris-setosa
8	4.6	3.4	1.4	0.3	Iris-setosa
9	5	3.4	1.5	0.2	Iris-setosa
10	4.4	2.9	1.4	0.2	Iris-setosa
11	4.9	3.1	1.5	0.1	Iris-setosa

Added Features: Flower ID, Sepal Area, and Petal Area

➔ Flower ID:

- Added an “Index” column as shown in below screenshot and renamed it as “Flower ID” column.

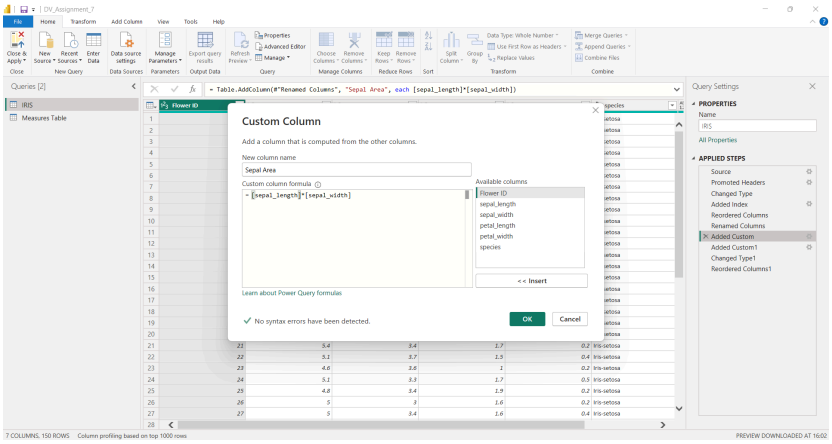




➔ Sepal Area:

- Go to ‘Add Column’ tab then click on the ‘Custom Column’ option.
- Created a Custom Column “Sepal Area” as shown in below screenshot using

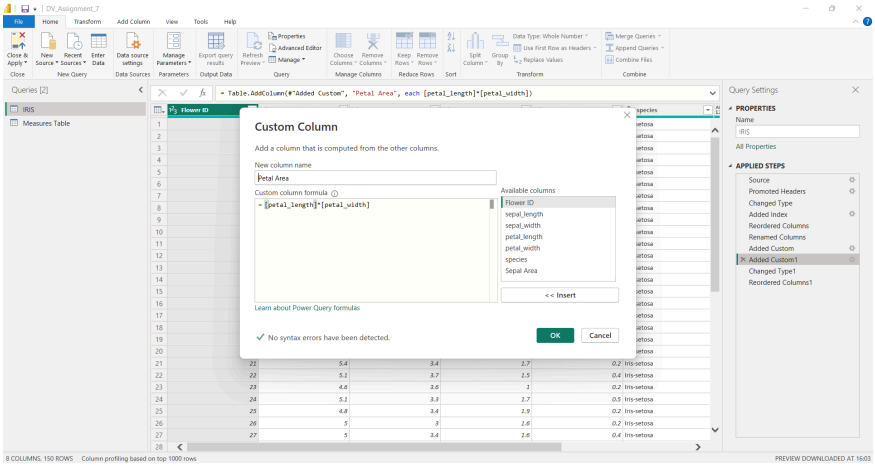
Sepal Area = Sepal Length x Sepal Width

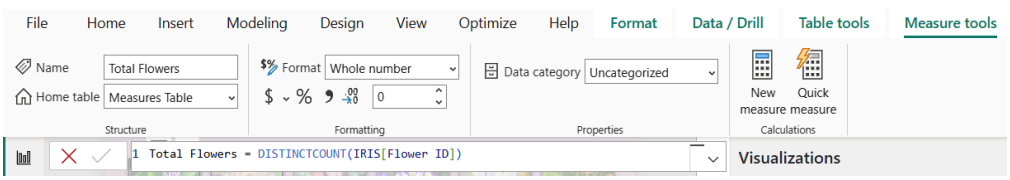


➔ Petal Area:

- Go to the ‘Add Column’ tab then click on the ‘Custom Column’ option.
- Created a Custom Column “Petal Area” as shown in below screenshot using

  Petal Area = Petal Length x Petal Width



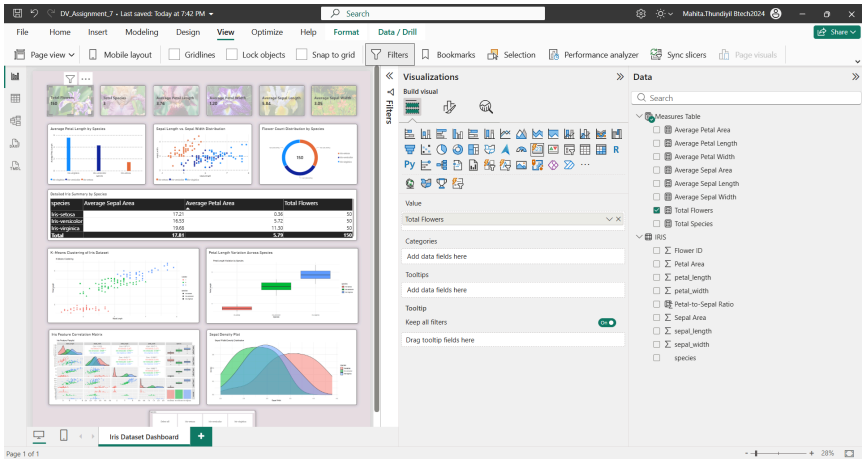


3. Key DAX Measures and Metrics

For the card tiles we will be using the following DAX measures:

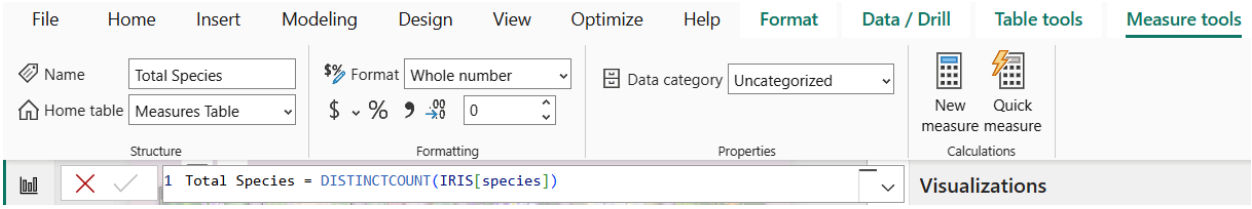
➔ Total Flowers = DISTINCTCOUNT (Iris[Flower ID])

Created its Visual by click on card and then dragged and dropped this measure in Visualisations Pane

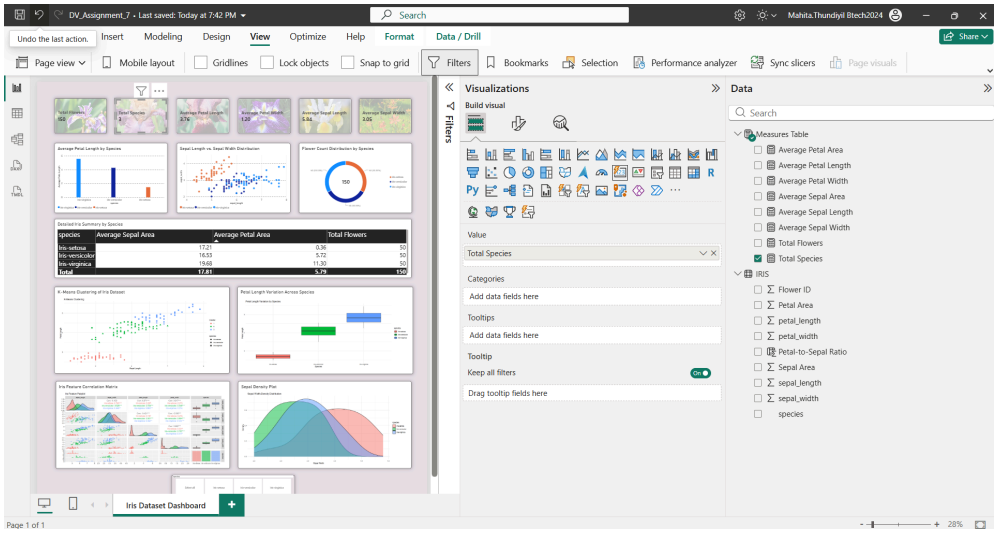




➔ Total Species = DISTINCTCOUNT (Iris[species])

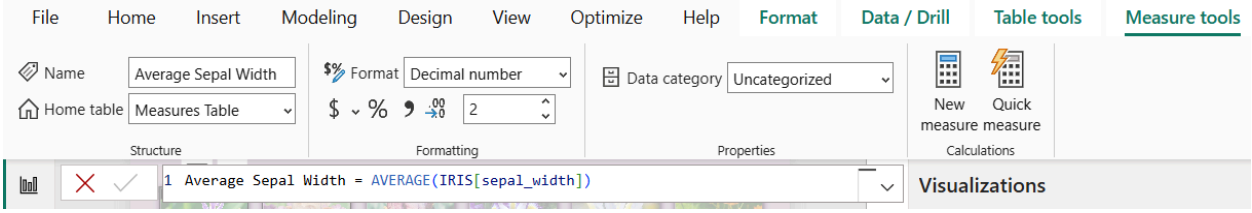


Created its Visual by click on card and then dragged and dropped this measure in Visualisations Pane

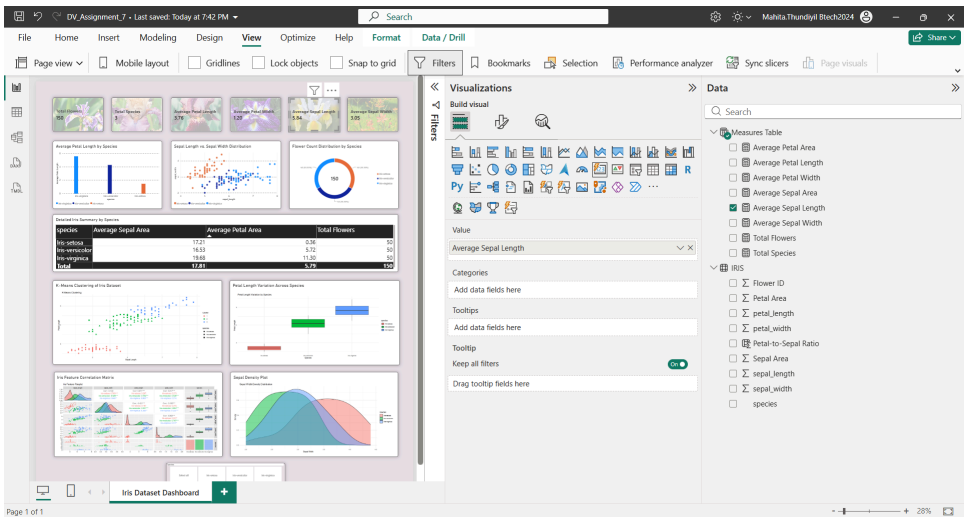


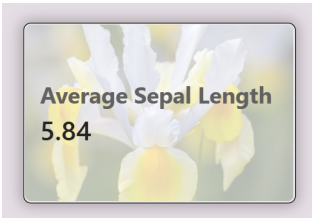


➔ Avg Sepal Length = AVERAGE (Iris[sepal_length])

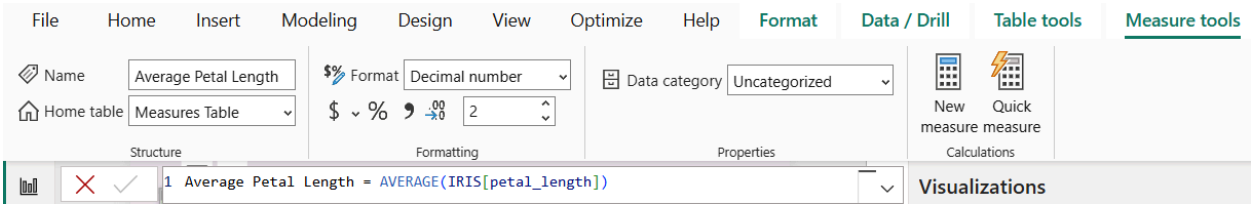


Created its Visual by click on card and then dragged and dropped this measure in Visualisations Pane

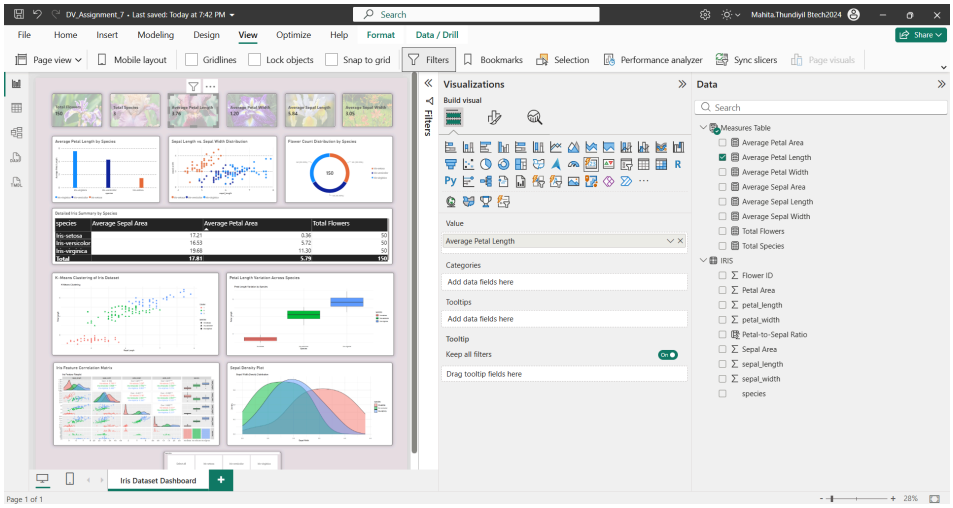


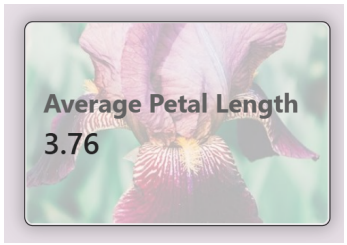


➔ Avg Petal Length = AVERAGE (Iris[petal_length])

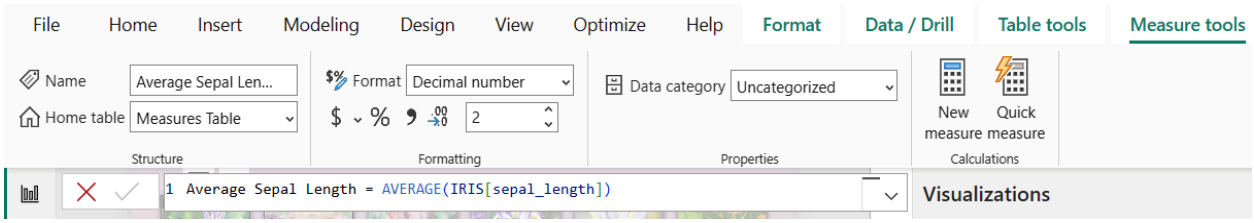


Created its Visual by click on card and then dragged and dropped this measure in Visualisations Pane

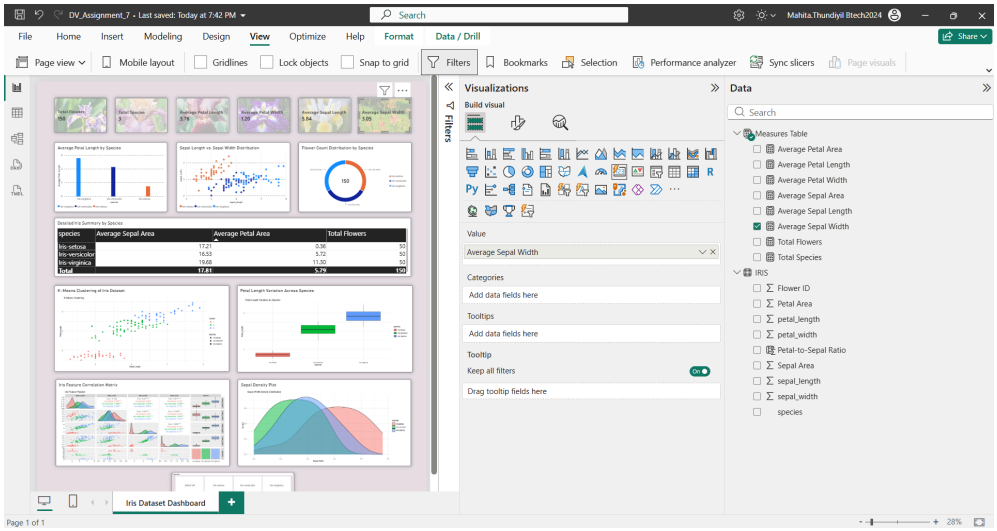




➔ Avg Sepal Width = AVERAGE (Iris[Sepal Area])

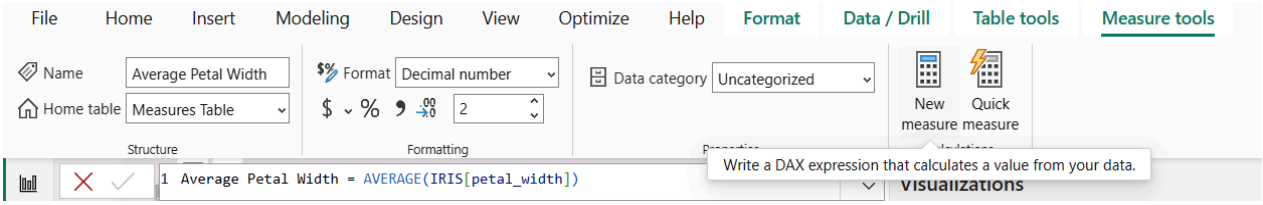


Created its Visual by click on card and then dragged and dropped this measure in Visualisations Pane

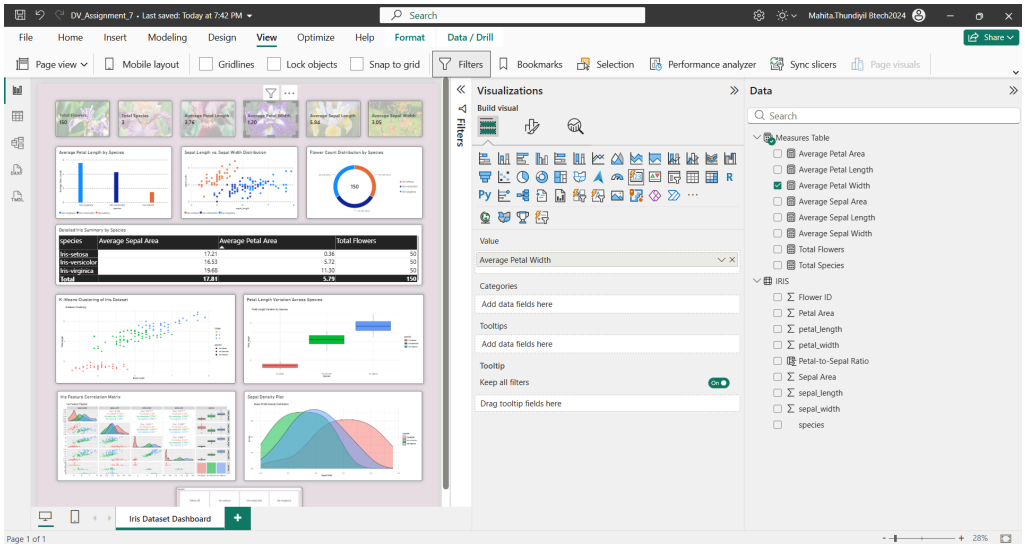




➔ Avg Petal Width = AVERAGE (Iris[Petal Area])



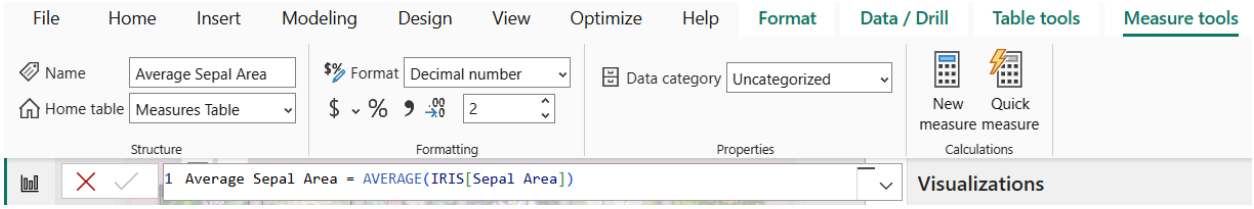
Created its Visual by click on card and then dragged and dropped this measure in Visualisations Pane



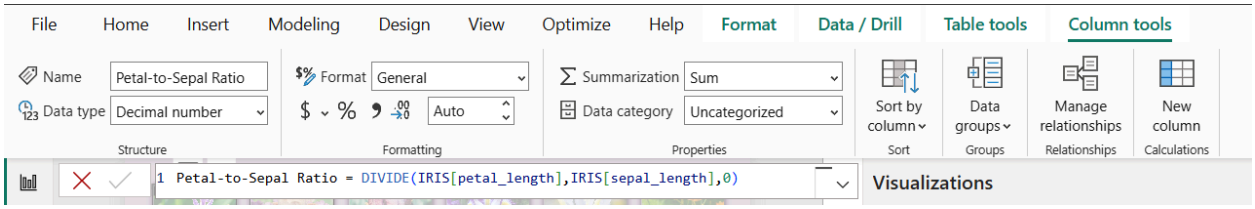


The other DAX measures that I created are:

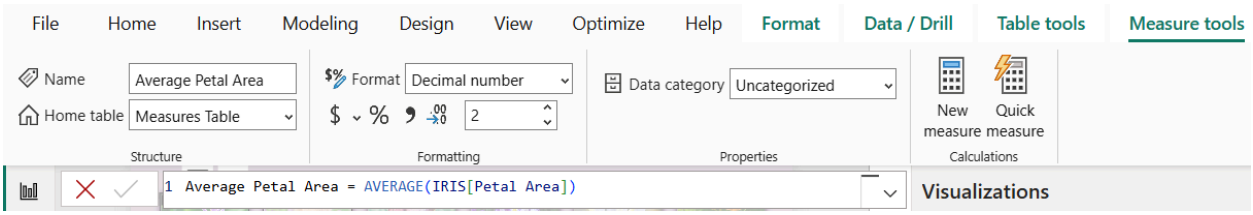
➔ Average Sepal Area = AVERAGE (Iris[Sepal Area])



➔ Petal-to-Sepal Ratio = DIVIDE (Iris[petal_legth], Iris[sepal_legth],0)

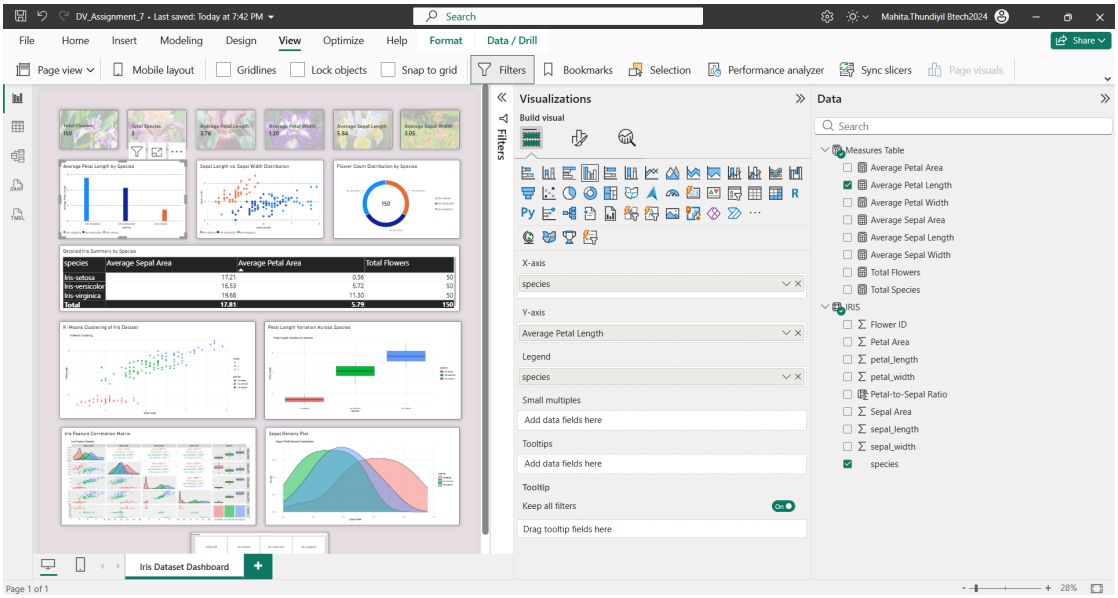


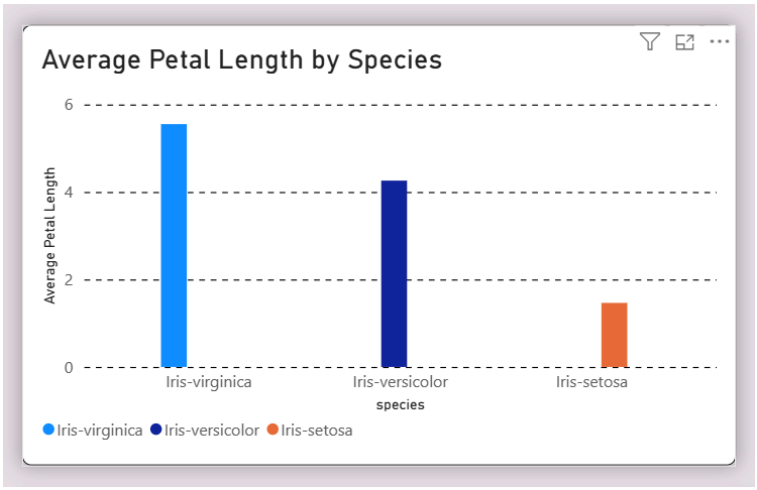
➔ Average Petal Area = AVERAGE (Iris[Petal Area])



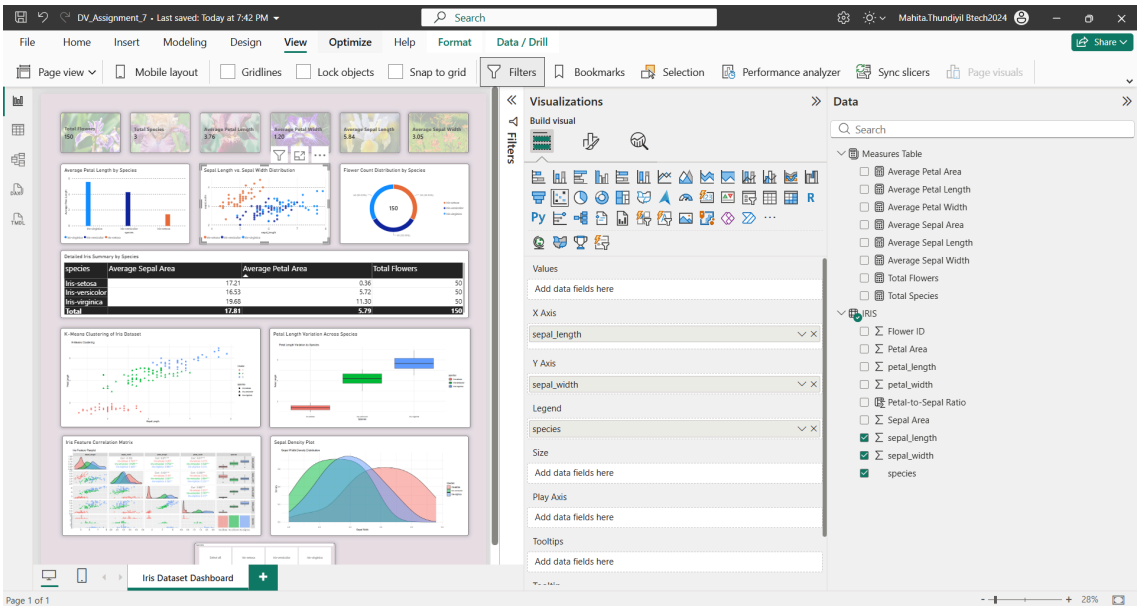
4. Visualisations and Dashboard

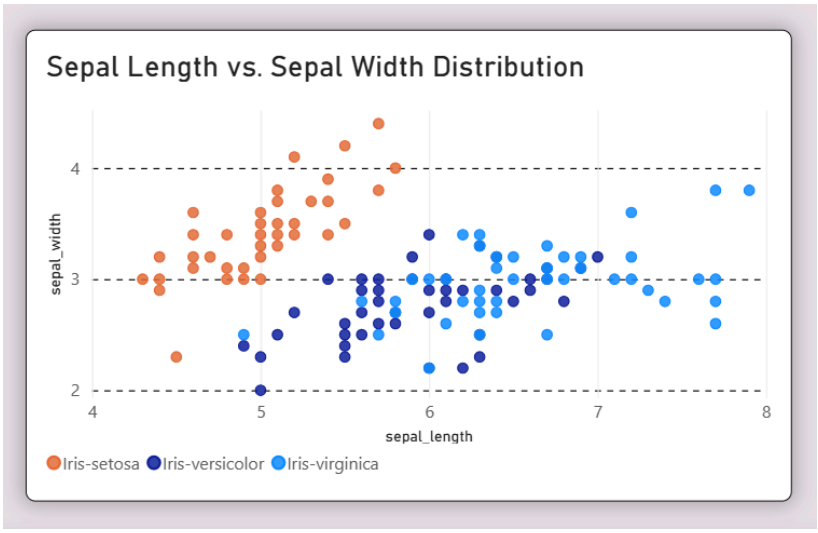
➔ I created a Clustered Column chart to show the relation between petal length and species by dragging and dropping the fields into a Clustered Column visual as shown in below screenshot:



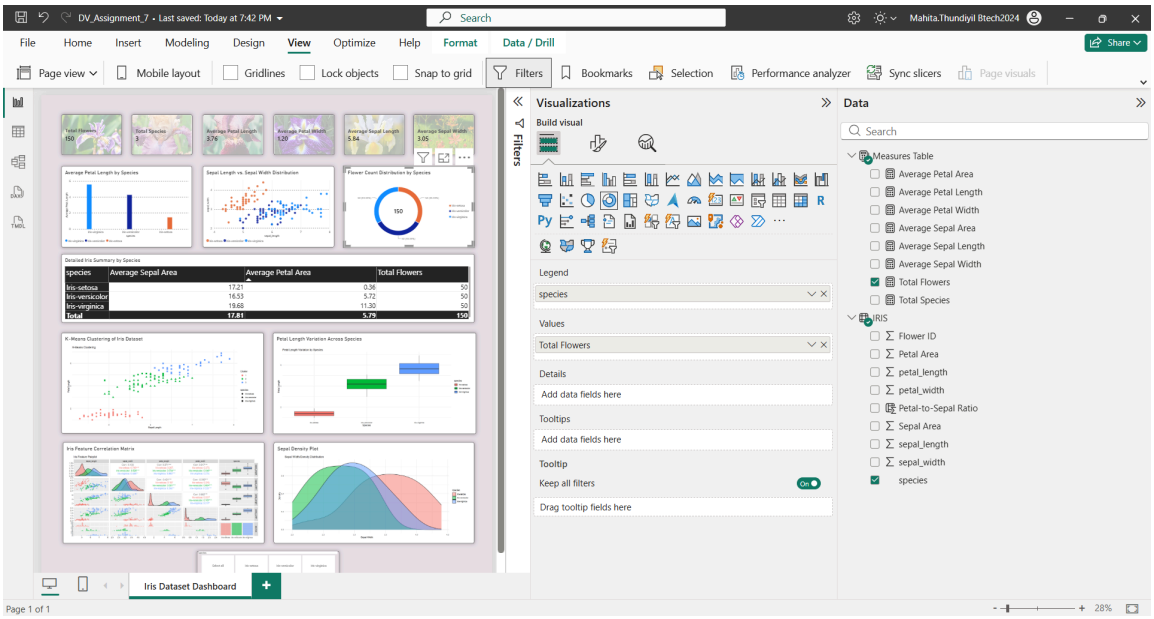


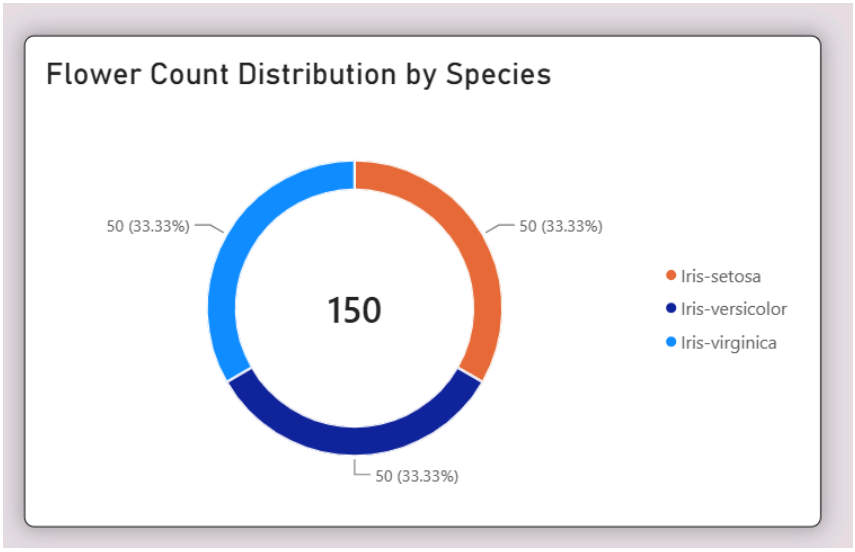
➔ I created a scatter plot to show the relation between sepal length and sepal width by dragging and dropping the fields into a Scatter Plot visual as shown in below screenshot:



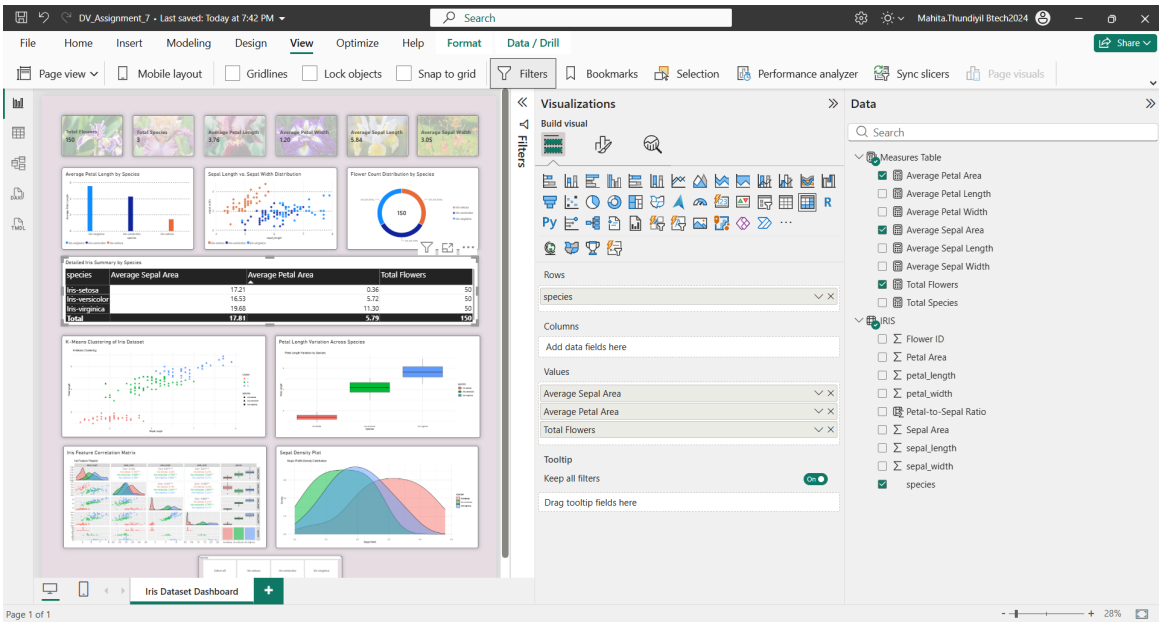


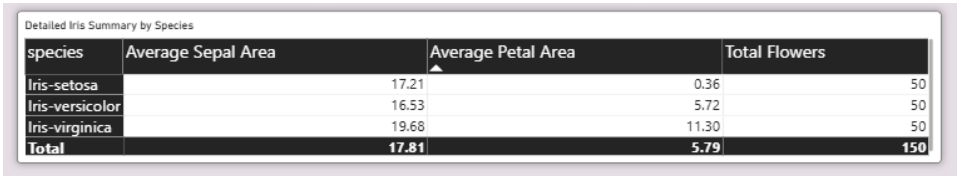
➔ I created Doughnut chart a to show the distribution of species across the records by dragging and dropping the fields into a Doughnut visual as shown in below screenshot:





➔ I created a Matrix by dragging and dropping the fields into a Matrix visual as shown in below screenshot:

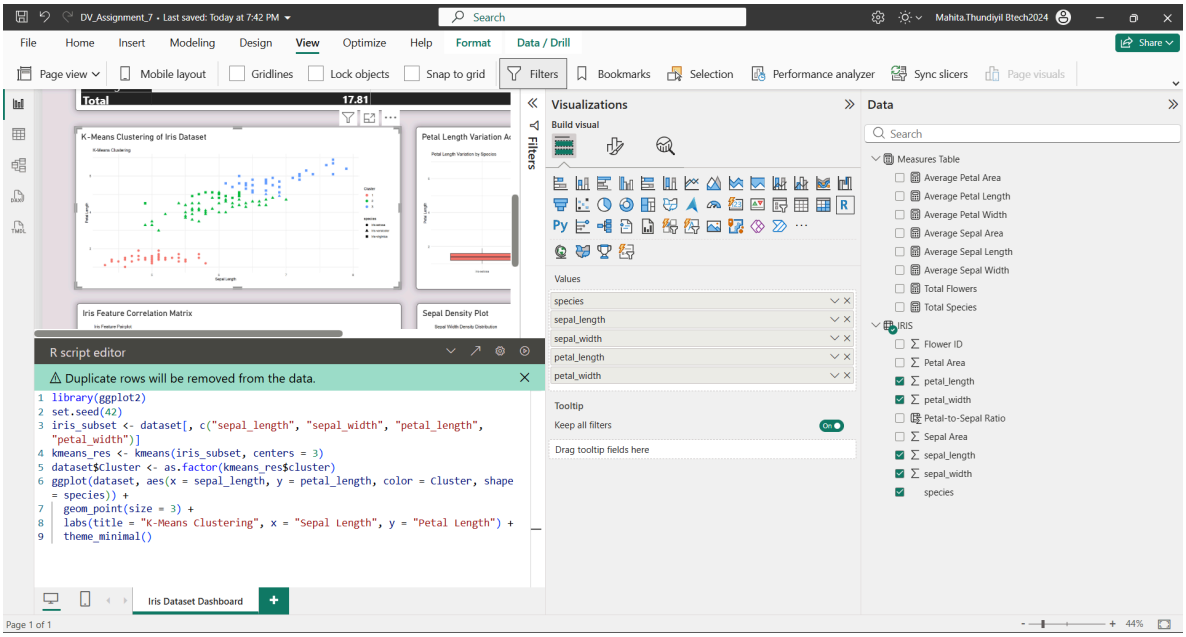


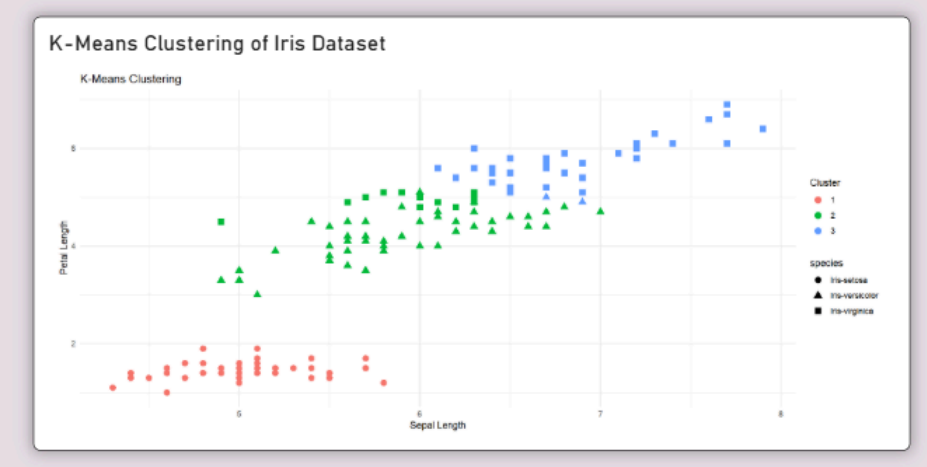


Detailed Iris Summary by Species

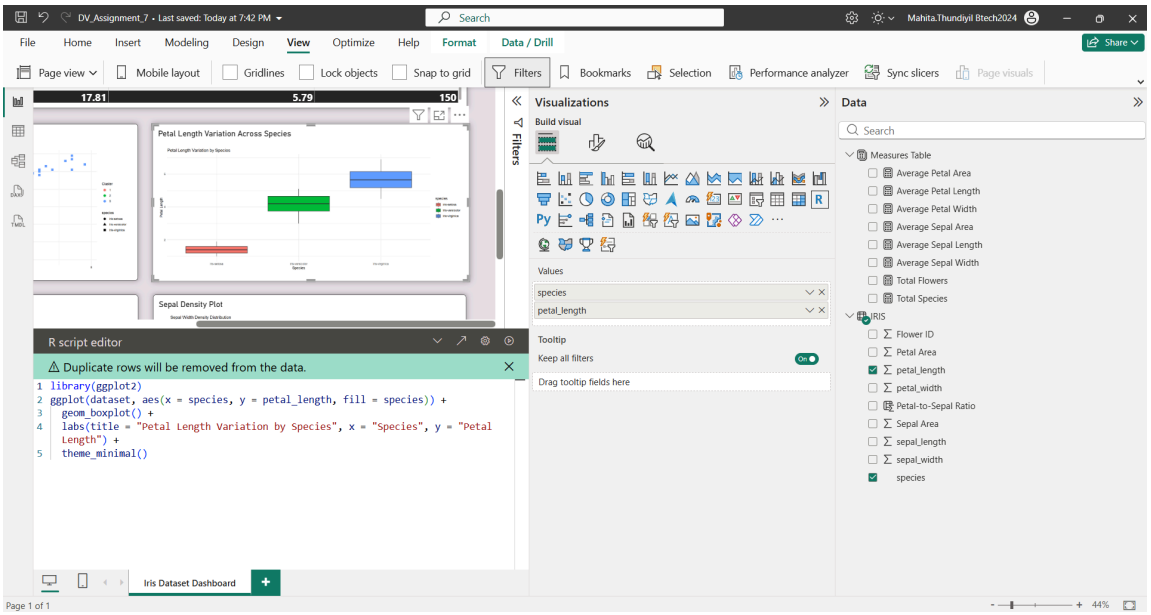
species	Average Sepal Area	Average Petal Area	Total Flowers
Iris-setosa	17.21	0.36	50
Iris-versicolor	16.53	5.72	50
Iris-virginica	19.68	11.30	50
Total	**17.81**	**5.79**	**150**

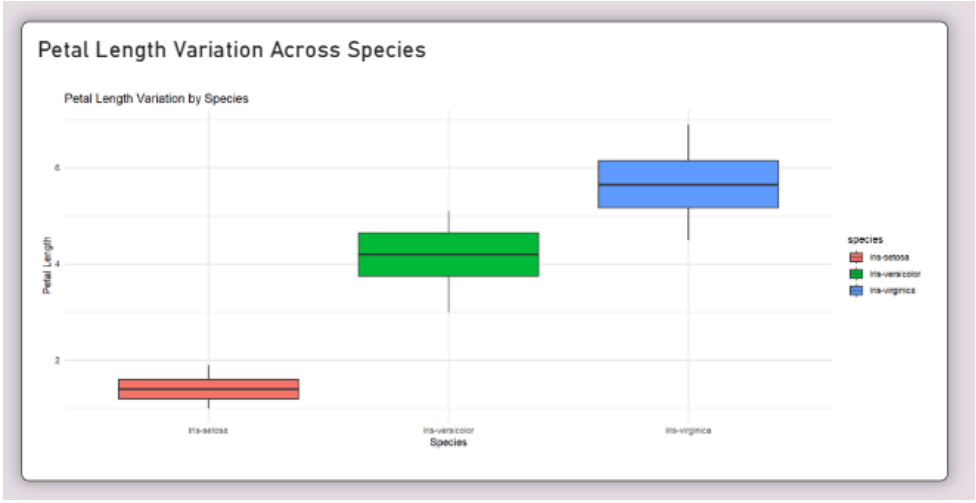
➔ I created a K-Means Clustering Plot through Power BI's integration with R. I first added the fields then I wrote the code as shown in below screenshot:



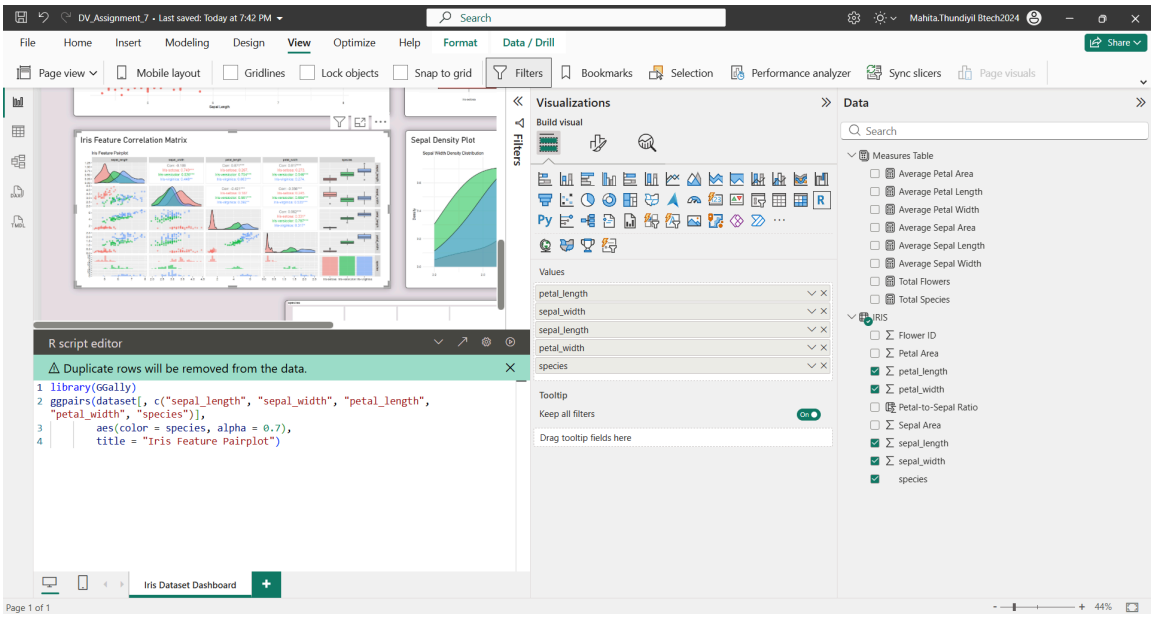


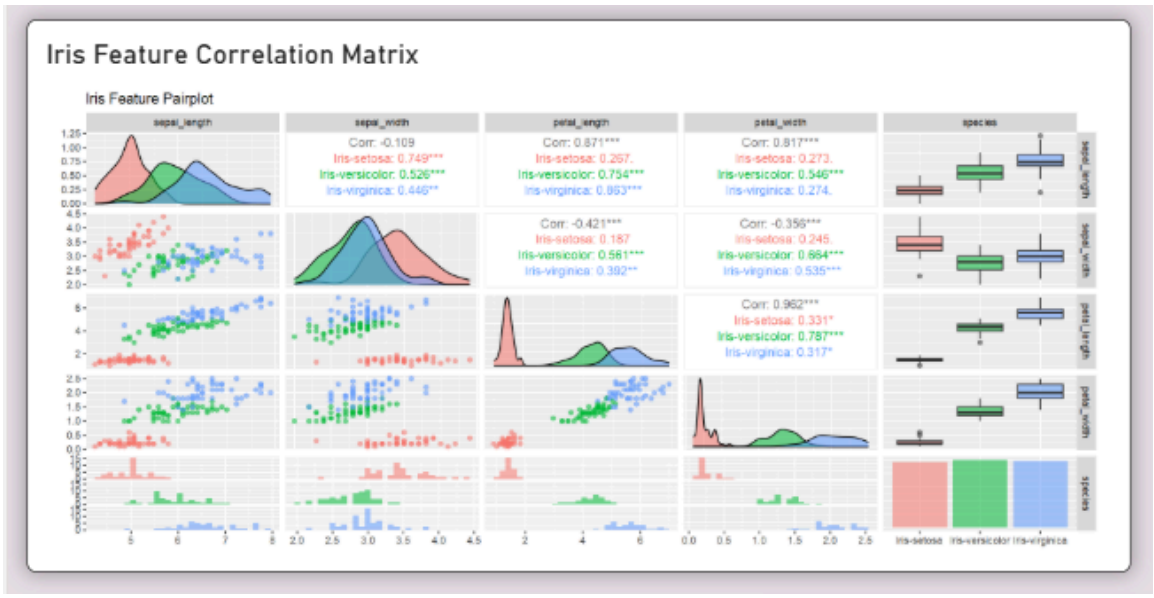
➔ I created a Box Plot through Power BI's integration with R. I first added the fields then I wrote the code as shown in below screenshot:



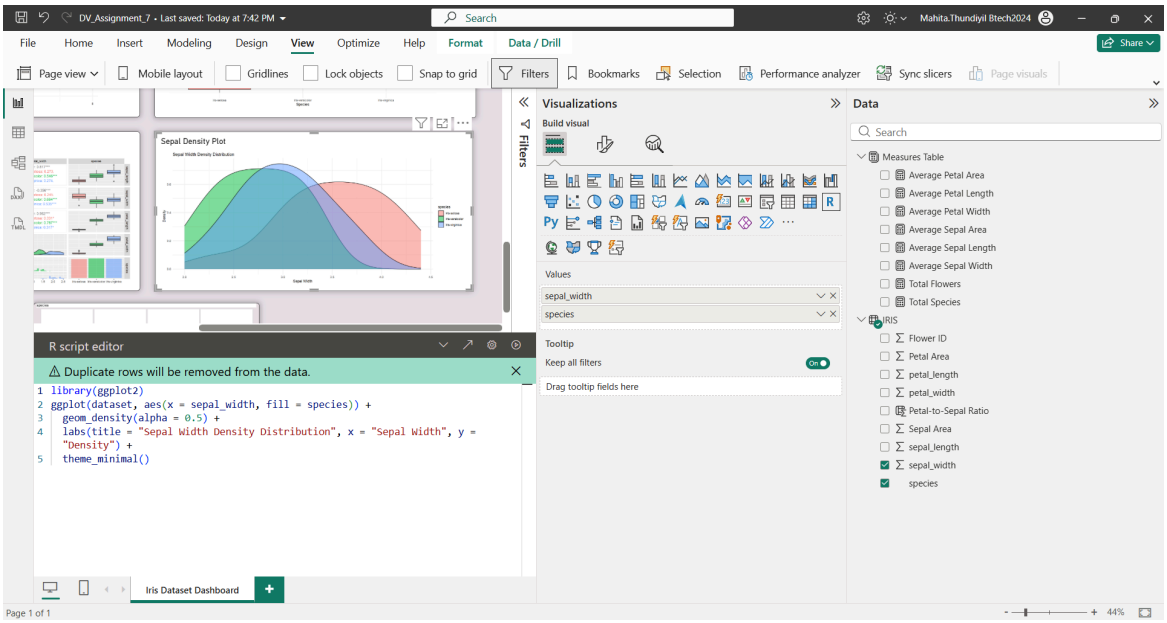


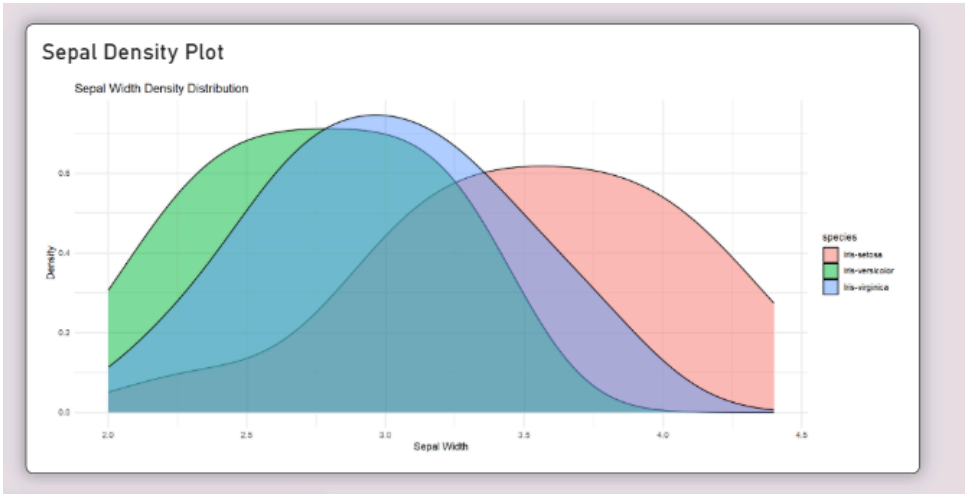
➔ I created a Feature Correlation through Power BI's integration with R. I first added the fields then I wrote the code as shown in below screenshot:



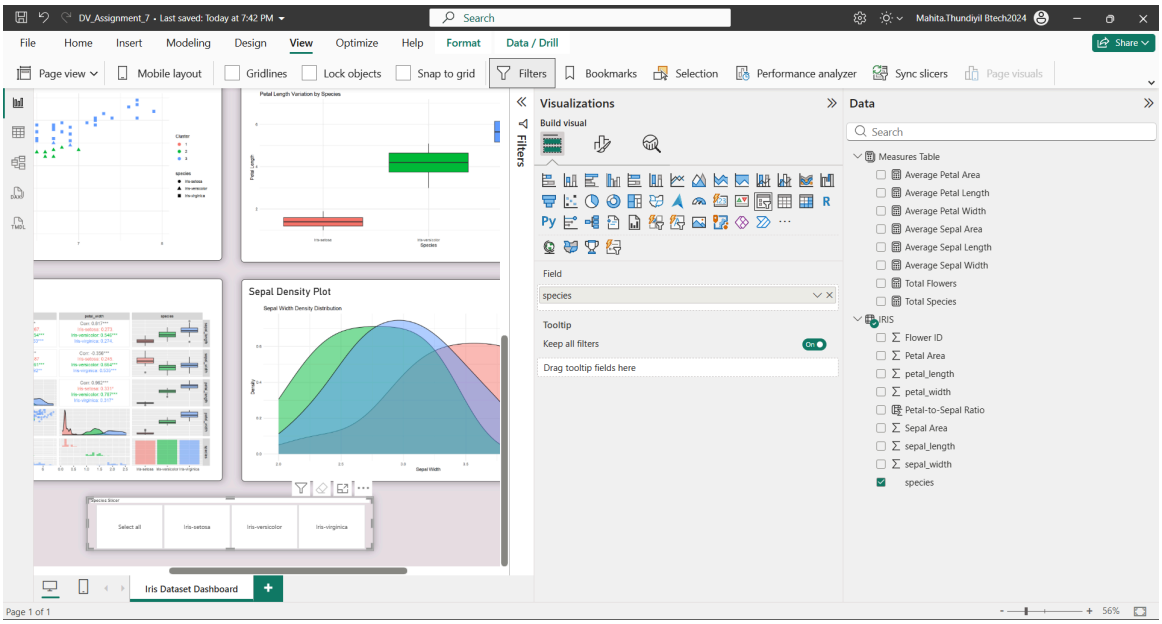


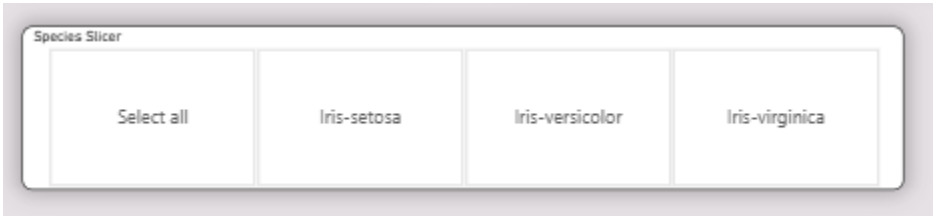
➔ I created a Sepal Density Plot through Power BI's integration with R. I first added the fields then I wrote the code as shown in below screenshot:



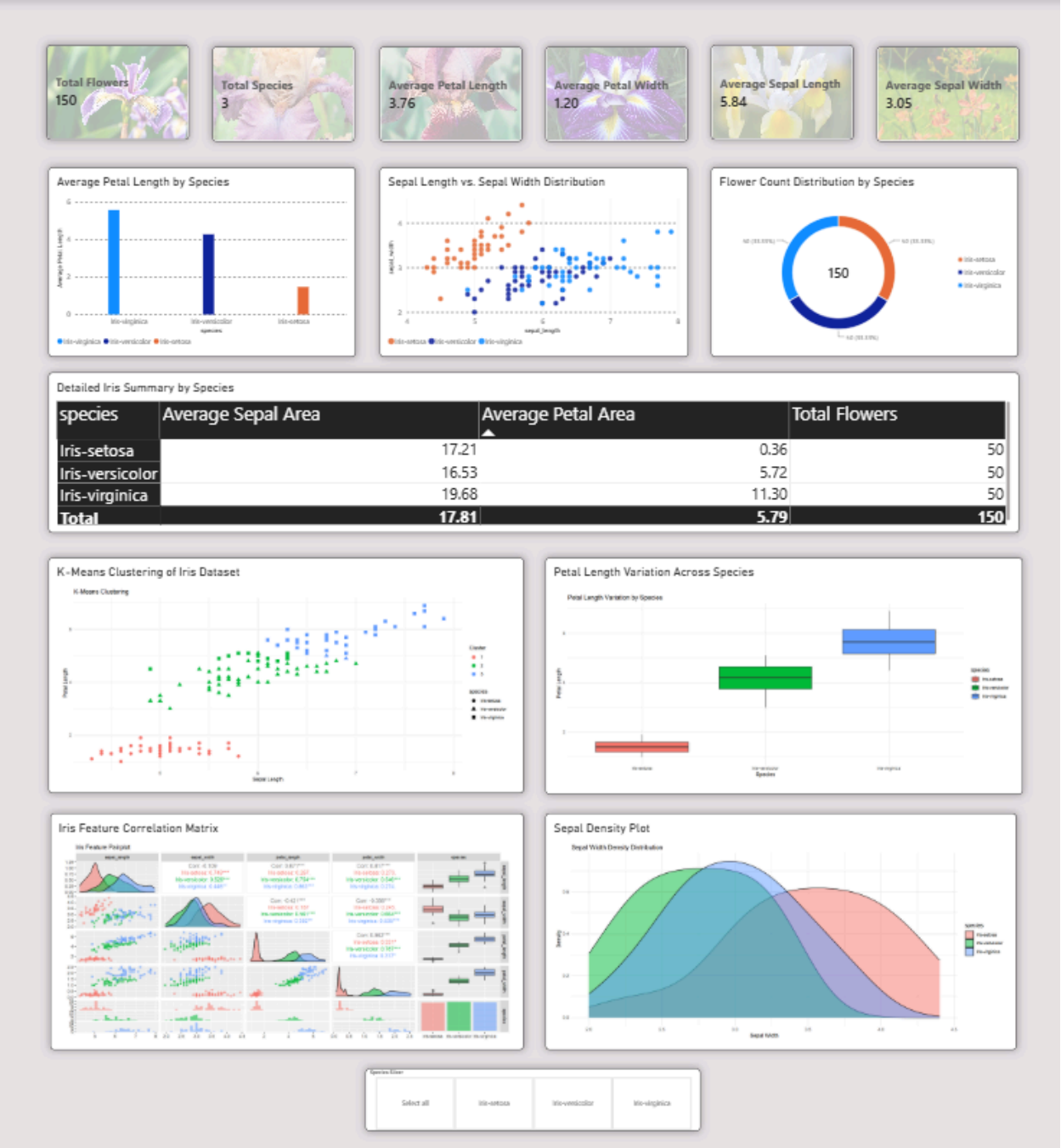


➔ I also added a “Species” slicer by dragging and dropping “species” into a Slicer visual as shown in below screenshot:





➔ Here's the overall view of my dashboard:



5. Conclusion

Creating the dashboard in Power BI allows us to visualize how our physical measurements contribute to the overall classification of the Iris flower. For example, we can see how the setosa species differ from the other species in terms of petal length, width, and surface area. Overall, this is a fun way to perform some basic data science while also learning the capabilities of Power BI and R.

Links to Dashboard

Platform	**Link**
Google Documents	https://docs.google.com/document/d/16s_7ZwaNuMfvBAcy9IGLrh_MCYu7zMSOn4ZodOe5N04/edit?usp=sharing
LinkedIn	https://lnkd.in/p/d93vn_Ya
GitHub (blog + PBIX file)	https://github.com/mst-2005/DataVisualization/tree/main/Experiment7